Singly Linked List – GetNthNodeFromEnd2 and Time Complexity.

[*Nth Node from end of a Linked List*, 0 *based indexing*]

This function helps us to get the last node from the end of linked list.

Program :

```
int getNthNodeFromEnd2(Node **head, int node)
{
    int len = 0;
    Node *current;
    current = *head;
    while (current != nullptr)
    {
        current = current->next;
        len++;
    }
    if (node < 0 || node >= len)
    {
        cout << "Error: Invalid index." << endl;
        return -1;
    }
    else
    {
        current = *head;
        for (int i = 0; i < len - node - 1; i++)
        {
            current = current->next;
        }
        return current->data;
    }
}
```

```
case 21:
        {
            cout << endl;
            cout << endl;
            cout << "Nth node from end , where 0 is the last node(0 based
indexing):" << endl;
            int n;
            cout << "Enter the value of node: ";
            cin >> n;
            cout << "The " << n << "th node from the end is: " <<
getNthNodeFromEnd2(&head, n) << endl;
            cout << endl;
            cout << endl;
            break;
        }
```

Program Analysis:

Function Call:

$Function\ call\ occurs\ when\ Case\ 21\ is\ initiated\ from\ Switch - Case$:

$cout \ldots\ldots getNthNodeFromEnd2(\&head, n);$

$There\ are\ two\ criteria\ of\ input\ Linked\ List$:

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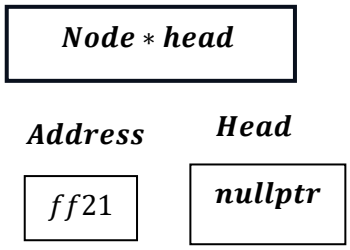
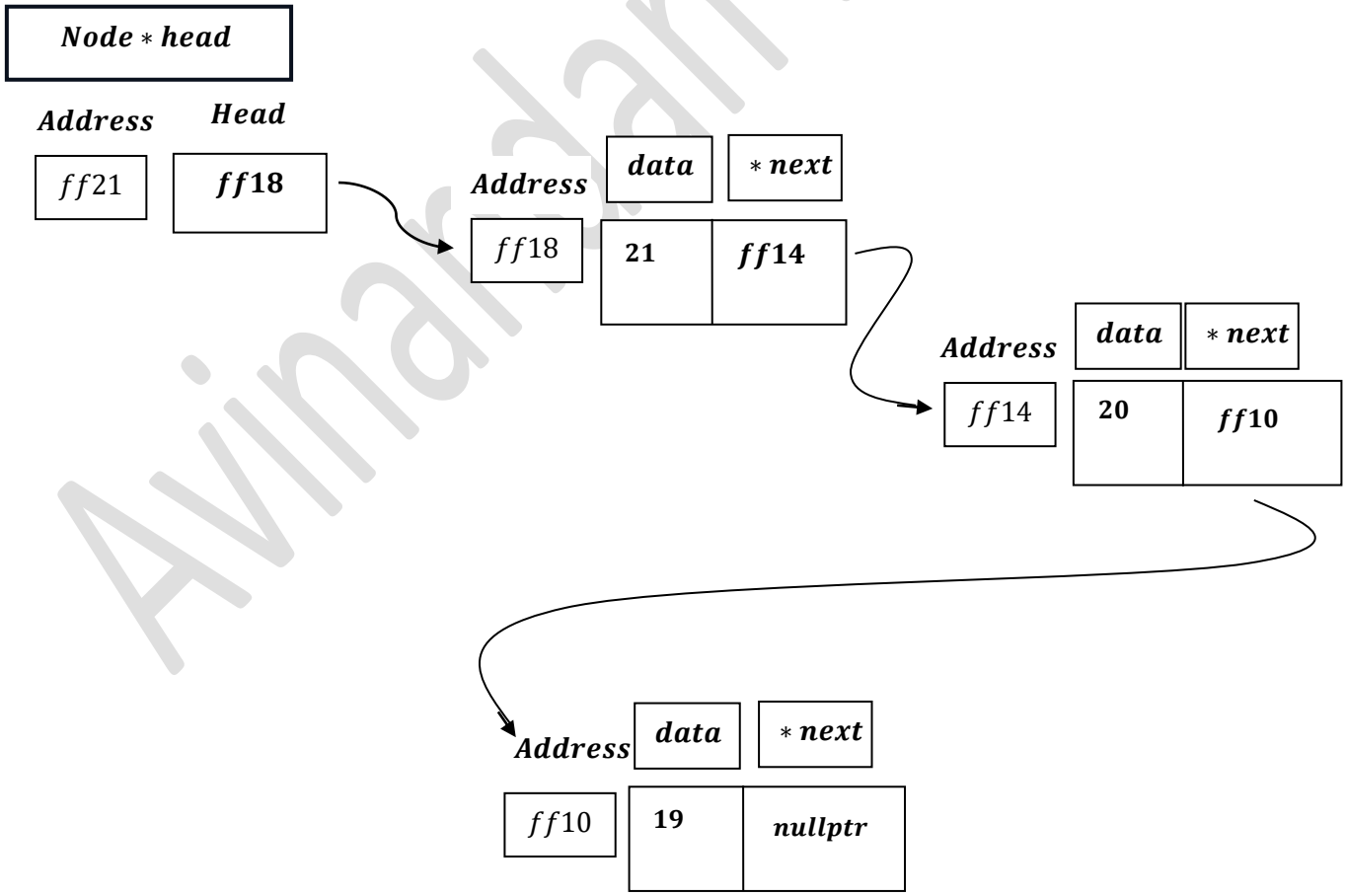
1) *Linked List is Empty*:

Pictorially depicted as:



2) *Linked List is not Empty*:

Pictorially depicted as:



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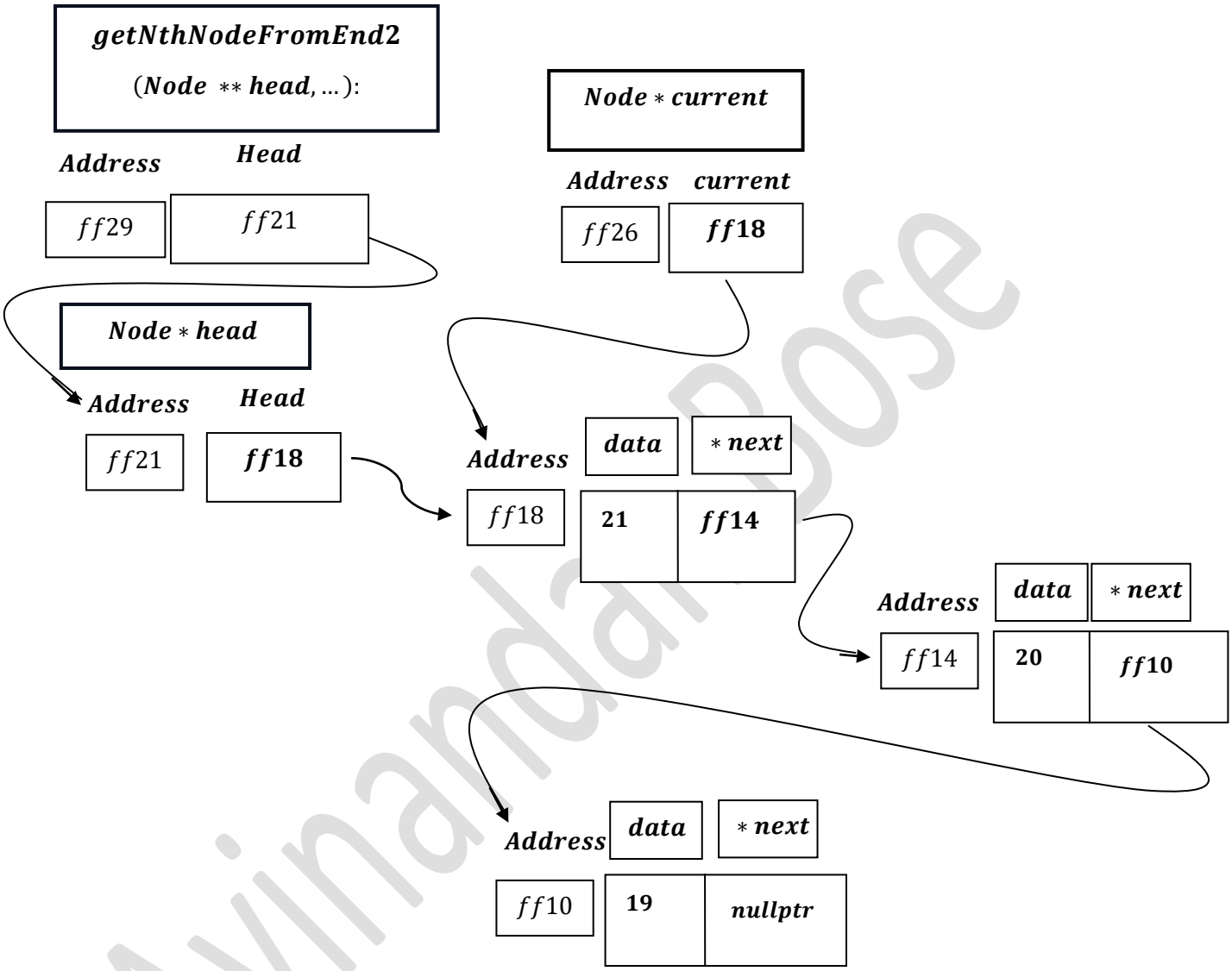
```
Node *current;
current = *head;
```




```
    int len = 0;
```

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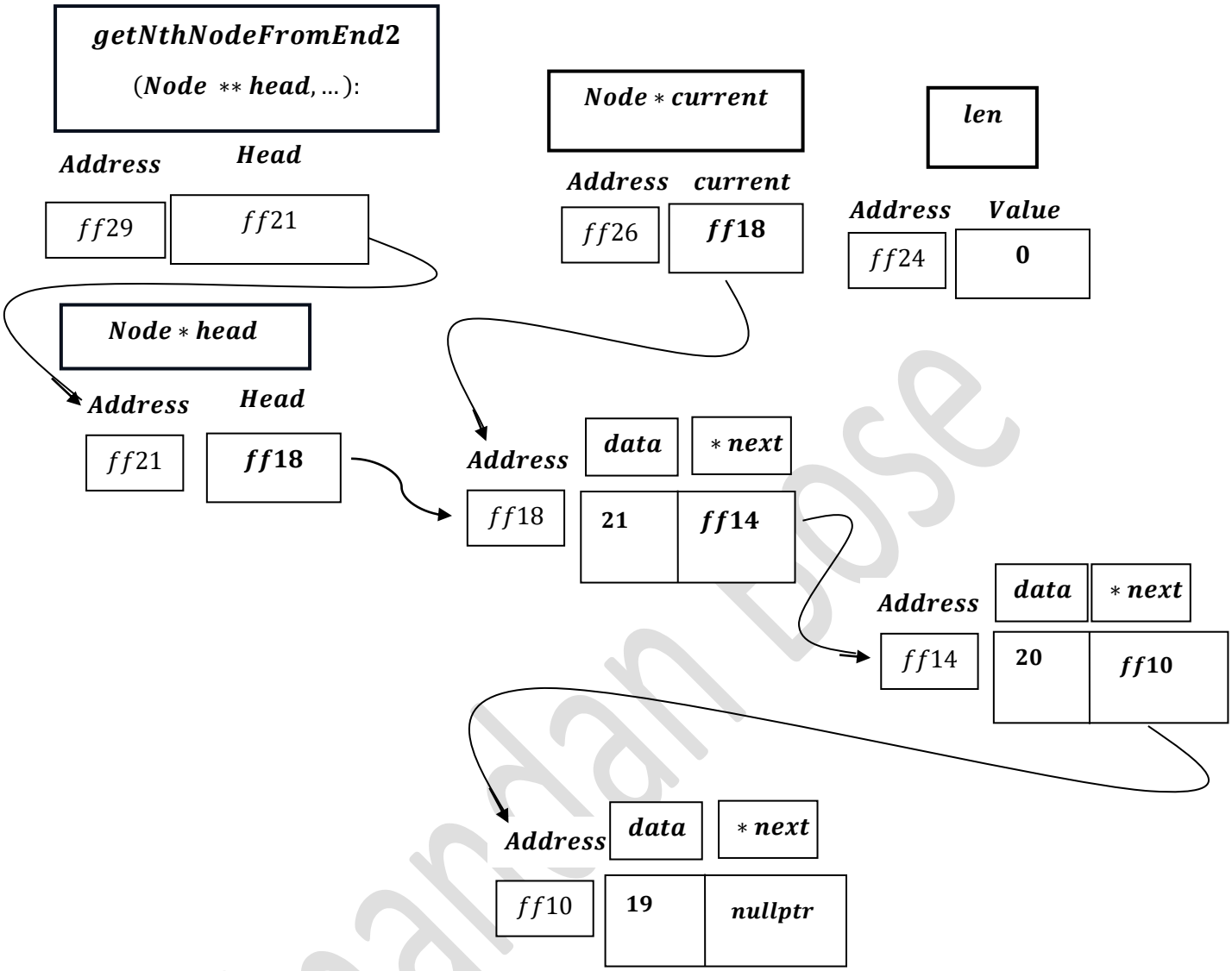
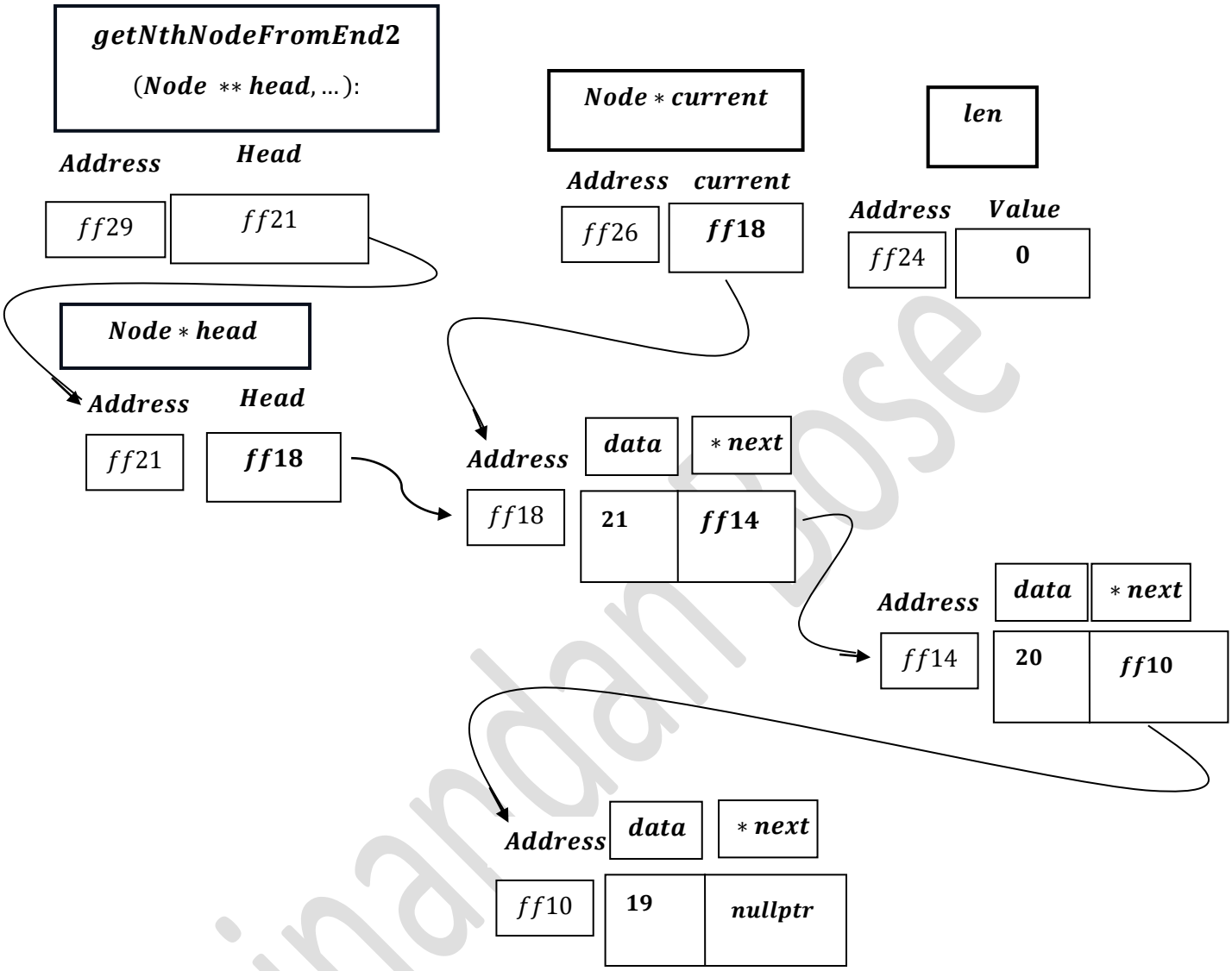


***Now , we will count length of the linked list*: –**

```
    while (current != nullptr)
    {
        current = current->next;
        len++;
    }
```

***We just recall the above picture* :**




```
while(current != nullptr){
        current = current -> next; i.e. current = ff14;
        len ++; i.e. len = 0 + 1 = 1;
}
```

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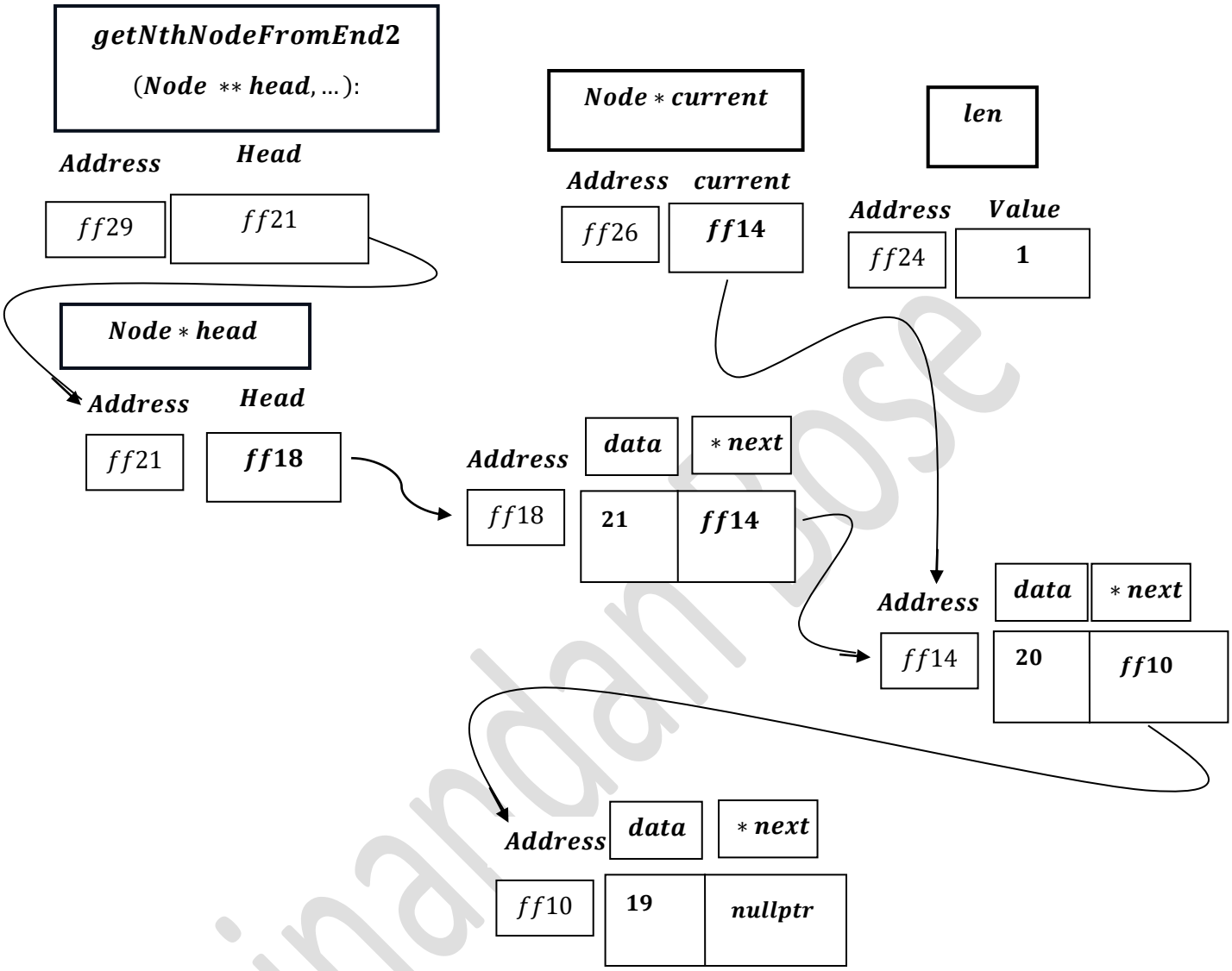


```
while(current != nullptr){
        current = current -> next; i.e. current = ff10;
        len ++; i.e. len = 1 + 1 = 2;
}
```

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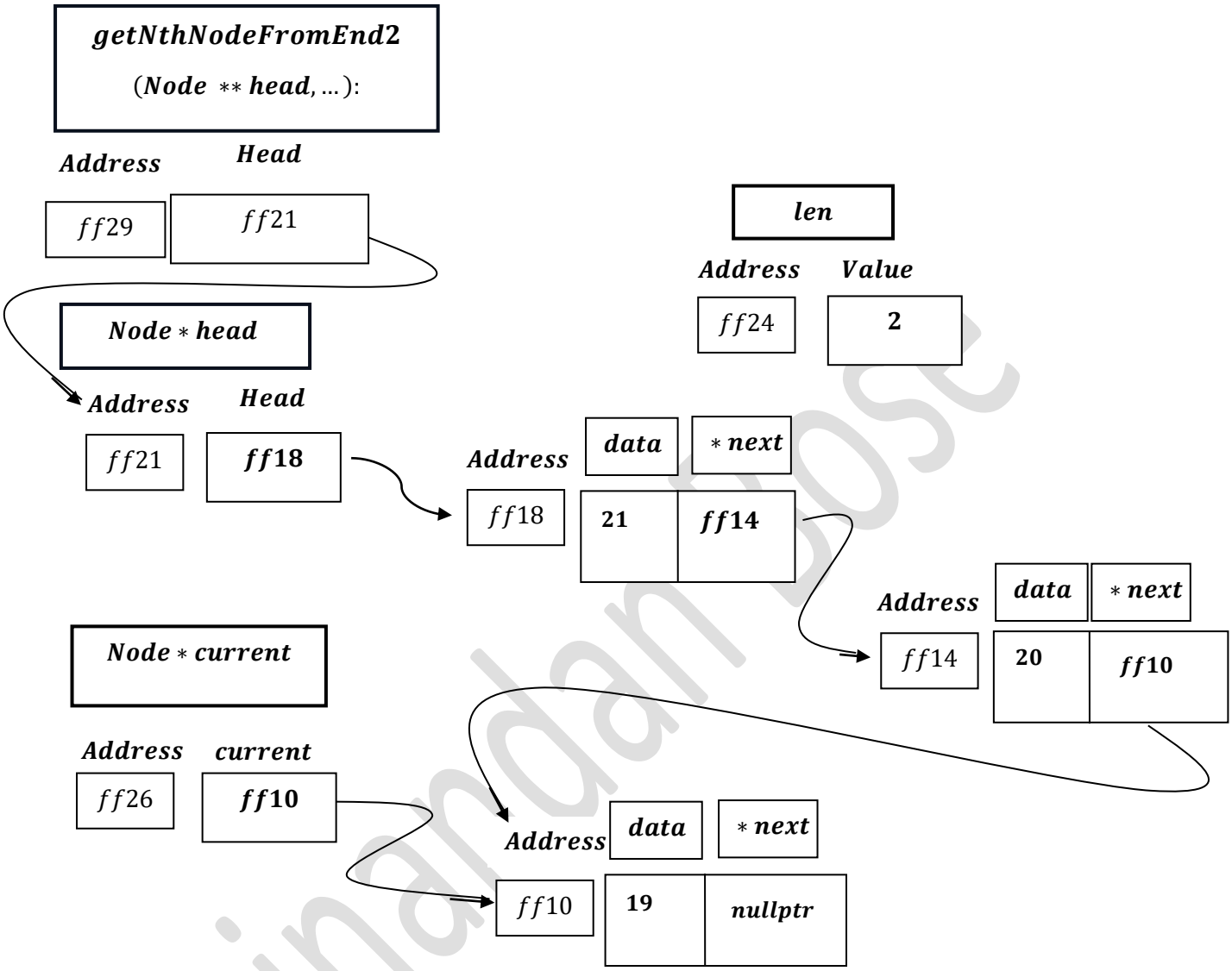


```
while(current != nullptr){
        current = current -> next; i.e. current = nullptr;
        len ++; i.e. len = 2 + 1 = 3;
}
```

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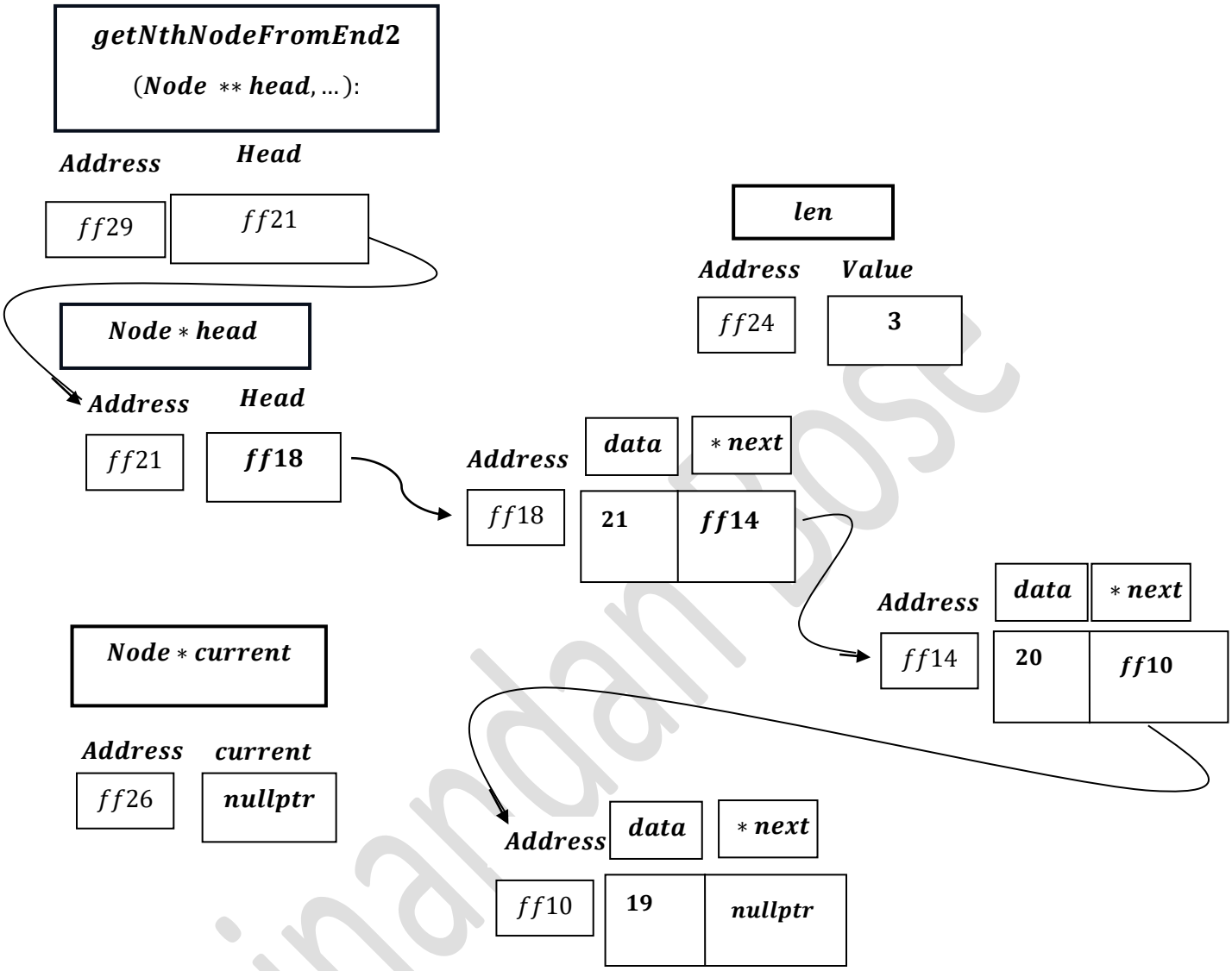


Now, it checks whether user input `node` is less than `0` or `node` greater than or equal to `lengthOfLinkedList` as shown below:

```
if (node < 0 || node >= len)
    {
        cout << "Error: Invalid index." << endl;
        return -1;
    }
```

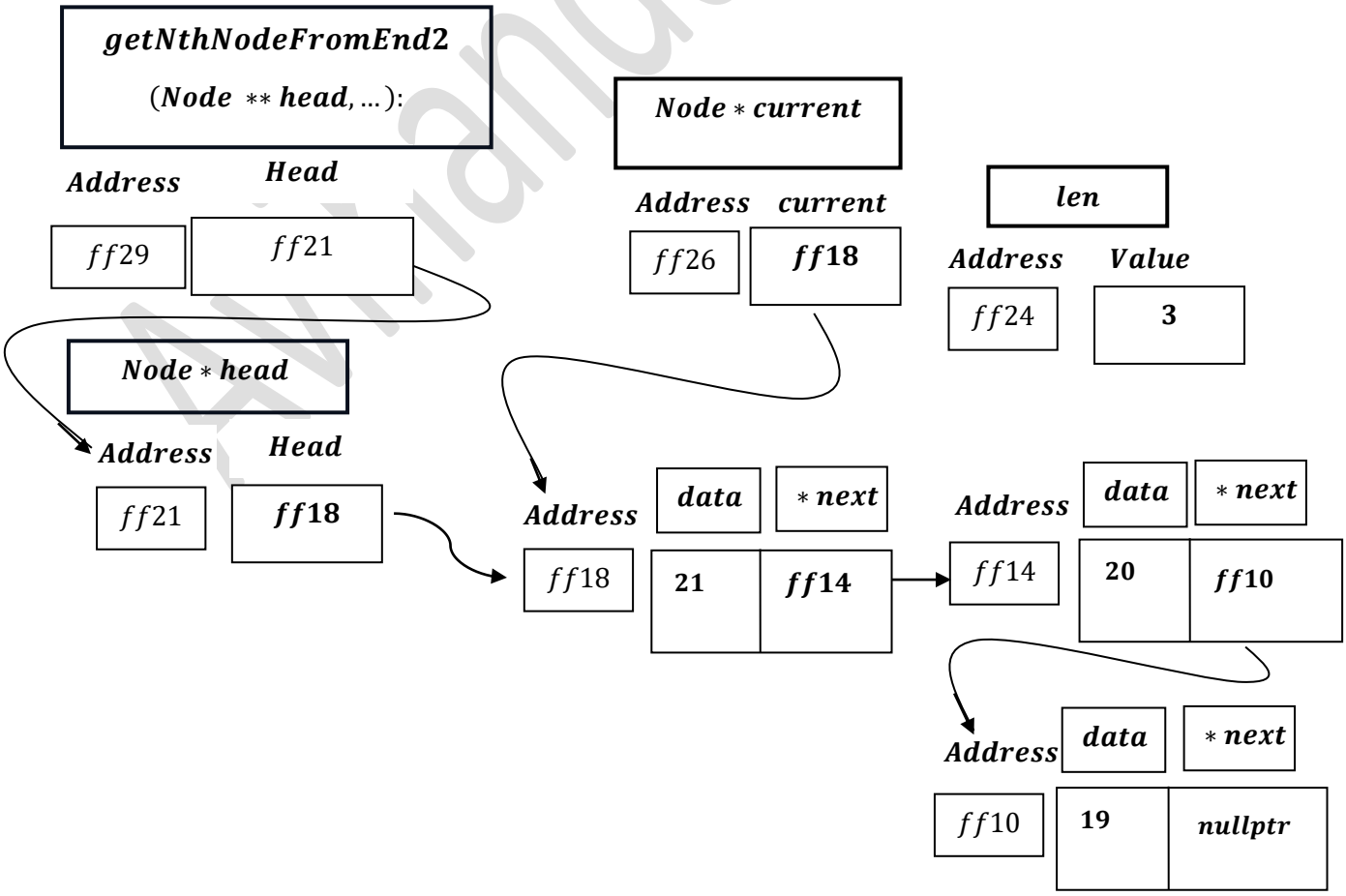
It outputs ``Error: Invalid index.`` and returns `−1`.

Now if user input node not less than 0 or `node` not greater than or equal to `lengthOfLinkedList` , then it will move to else part:

```
else
    {
        current = *head;
        for (int i = 0; i < len - node - 1; i++)
        {
            current = current->next;
        }
        return current->data;
    }
}
```

Now,

```
current = *head;
```



$Now,$

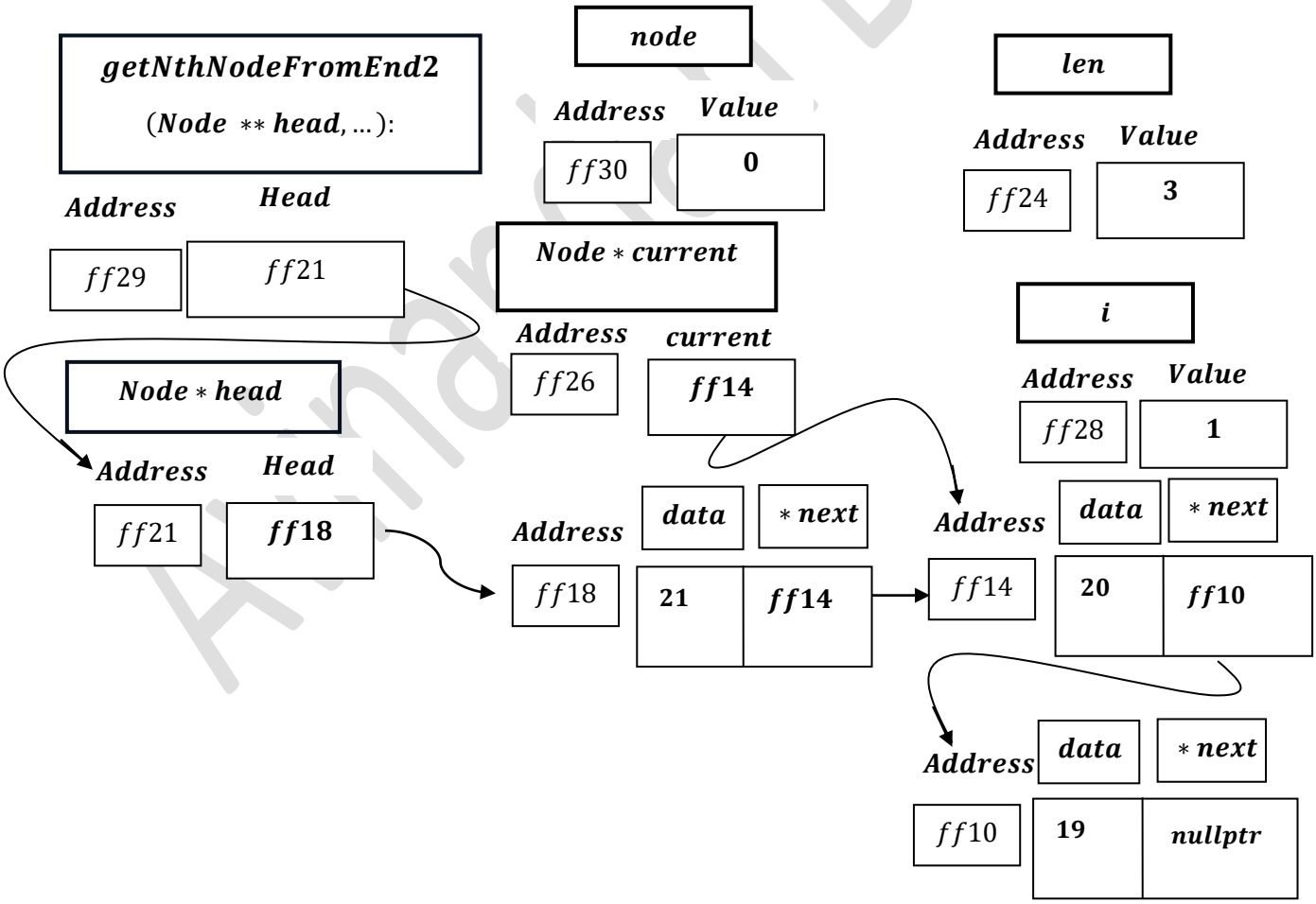
```
        for (int i = 0; i < len - node - 1; i++)
        {
            current = current->next;
        }
```

$Say, node = 0 (user\ input).$

$for\ i = 0\ and\ i < (len: 3 - node: 0 - 1)\ i.e.\ i < 2:$

$current = current \rightarrow next: ff14;$

$i = i + 1 = 0 + 1 = 1;$



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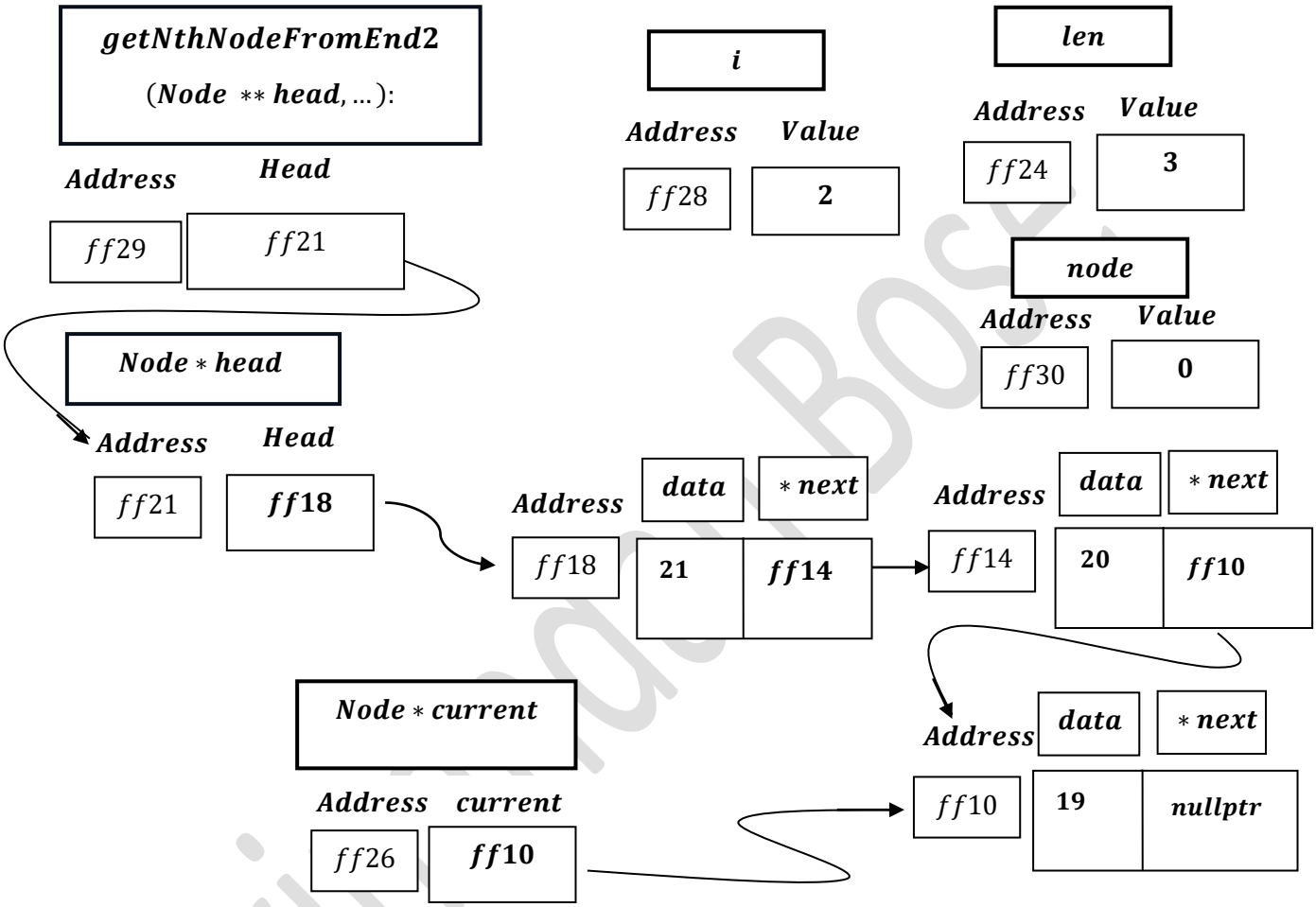
$for\ i = 1\ and\ i < (len: 3 - node: 0 - 1)\ i.e.\ i < 2:$

$current = current \rightarrow next: ff10;$

$i = i + 1 = 1 + 1 = 2;$



$For\ next\ updation, for\ i = 2\ and\ \ i < 2\ , hence\ \ condition\ is\ false\ ,$

$hence\ loop\ exits.$

```
        return current->data;
```

$Now, it\ returns\ current \rightarrow data; i.e, data\ stored\ in\ 3rd\ Node\ i.e.\ 19\ .$

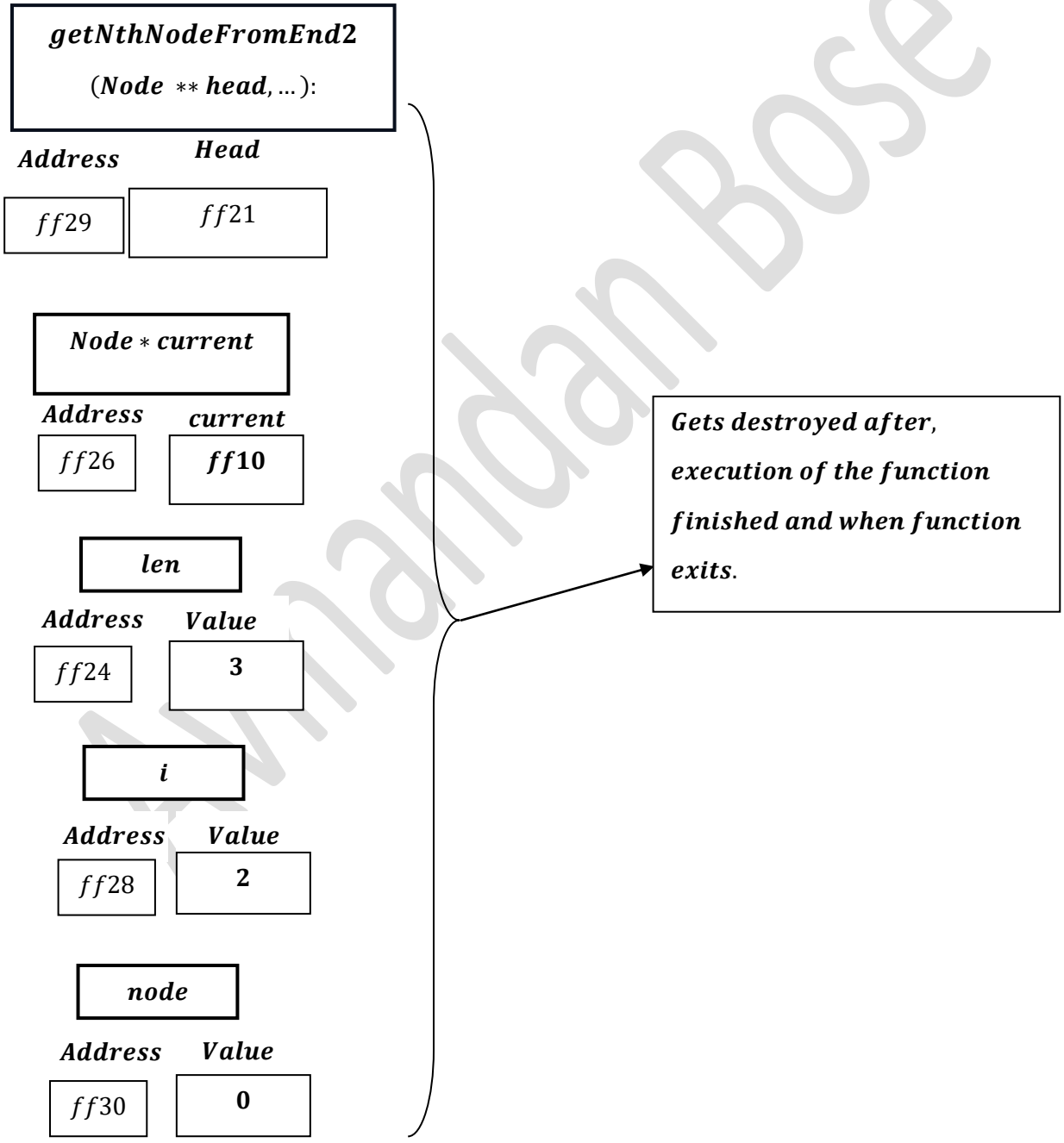
Thus we get Node from End i.e. if we enter node: 0 → It will give us n^{th} node's data.

if we enter node: 1 → It will give us $(n-1)^{th}$ node's data.

if we enter node: 2 → It will give us $(n-2)^{th}$ node's data.

.... And it goes like this.

Now, as the function get finished, all temporary variable created earlier in process for the program will get destroyed.



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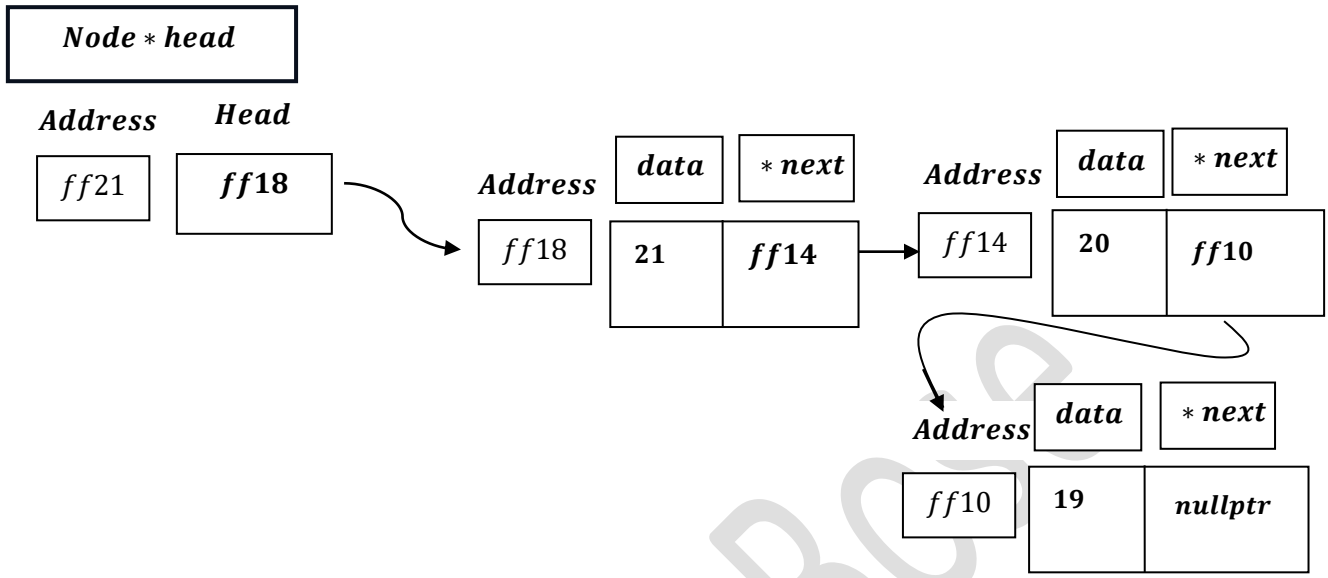
And we are left with:



Now, `getNthNodeFromEnd2` step by step explanation :

1. *Initialize Length and Current Pointer*:

- *An integer len is initialized to* 0 *to keep track of the length of the linked list.*
- *A pointer current is initialized to point to the head of the list.*

2. *Calculate the Length of the List*:

- *The function enters a loop to traverse the entire list to calculate its length.*
- *In each iteration, it moves the current pointer to the next node and increments len.*

3. *Check for Invalid Node Index*:

- *The function checks if the provided node index is less than* 0 *or greater than or equal to the length of the list.*

- *If the index is invalid, it prints an error message and returns* -1.

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4. *Reset Current Pointer*:

- *The current pointer is reset to point to the head of the list again.*

5. *Traverse to the* $(len - node - 1)th$ *Node*:

- *The function enters a loop to traverse the list until it reaches the* $(len - node - 1)th$ *node from the start.*

  $for\ (int\ i = 0;\ i < len - node - 1;\ i++)$: *This loop moves current to the desired node. The logic* $len - node - 1$ *is used because we are looking for the nth node from the end.*

- *This is equivalent to the nth node from the end of the list in a* $0-$*based index system.*

6. *Return Node Data*:

- *Once the desired node is reached, the function returns the data of the current node.*

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Time Complexity of getNthNodeFromEnd

```
int getNthNodeFromEnd(Node **head, int node)
{
    int len = 0;                                  O(1)
    Node *current;                                O(1)
    current = *head;                              O(1)
    while (current != nullptr)
    {
        current = current->next;                  runs n times.
        len++;                                    runs n times.
    }
    if (node < 0 || node >= len)                  checks condition in constant time.
    {
        cout << "Error: Invalid index." << endl;  runs 1 time i.e. O(1)
        return -1;
    }

    else
    {
        current = *head;                          takes O(1) time.
        for (int i = 0; i < len - node - 1; i++)
        {
            current = current->next;              if node is 0, loop runs `n − 1` times,
                                                  statement also runs `n − 1` times.
        }
        return current->data;                     takes O(1) time.
    }
}
```

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$\underline{Worst\ Case}$

$\underline{1.When\ Node\ is\ negative\ and\ node\ is\ greater\ than\ length:}$

- $when\ `len = 0`\ , this\ assignment\ takes\ constant\ amount\ of\ time\ , O(1).$
- $Node * current\ ; declaration\ takes\ constant\ amount\ of\ time, O(1).$
- $current = * head; assignment\ takes\ constant\ amount\ of\ time\ , O(1).$
- $Calculation\ of\ length\ `len`\ takes\ `n`\ times\ also\ `current = current \rightarrow next`$ $assignment\ takes\ `n`\ times\ , hence\ it\ takes\ in\ total: n + n = 2n = O(n).$
- $As\ `node`\ is\ less\ than\ 0\ or\ `node`\ is\ greater\ than\ or\ equal\ to\ `len`\ , it\ will\ output$ $a\ statement\ and\ return - 1\ , which\ takes\ constant\ amount\ of\ time$ $, i.e. O(1).$

  $Hence\ total\ amount\ of\ time\ taken: -O(1) + O(1) + O(1) + O(n) + O(1) = O(n).$

$\underline{2.When\ `node`\ is\ entered : `0`}$

- $when\ `len = 0`\ , this\ assignment\ takes\ constant\ amount\ of\ time\ , O(1).$
- $Node * current\ ; declaration\ takes\ constant\ amount\ of\ time, O(1).$
- $current = * head; assignment\ takes\ constant\ amount\ of\ time\ , O(1).$
- $Calculation\ of\ length\ `len`\ takes\ `n`\ times\ also\ `current = current \rightarrow next`$ $assignment\ takes\ `n`\ times\ , hence\ it\ takes\ in\ total: n + n = 2n = O(n).$
- $Now\ , node\ is\ `0`\ , then\ it\ just\ checks\ that\ `node`\ less\ than\ `0`\ , which\ is\ false$ $this\ takes\ constant\ `c`\ amount\ of\ time\ which\ is\ `O(1)`\ .$
- $As\ `node`\ input\ is\ 0\ , therefore\ statement\ , `current = current \rightarrow next`\ ,$ $assignment\ takes\ `n - 0 - 1 = n - 1`\ times. As\ current\ already\ points\ to$ $head\ pointer\ or\ 1st\ node\ to\ start\ with\ and\ while\ loop\ starts\ it\ will\ get$ $assigned\ to\ address\ of\ 2nd\ node\ , then\ the\ 3rd\ node, \ldots, to\ address$ $of\ nth\ node. Hence\ 2nd\ to\ nth\ node\ i.e. loop\ runs\ n - 1\ times.$

  $Therefore\ , this\ results\ , O(n - 1) = O(n).$

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- $Now, it\ returns\ current \rightarrow data; which\ takes\ constant\ `c`\ amount\ of\ time,$ $which\ is: O(1).$

  $Hence\ total\ amount\ of\ time\ taken: -O(1) + O(1) + O(1) + O(n) + O(1) + O(n) + O(1) = O(n).$

$\underline{Average\ Case}$

$$\left(\begin{array}{c} uniform\ distribution\ of\ nodes - \\ distribution\ where\ all\ outcomes\ are\ equally\ likely. \end{array}\right)$$

$Uniform\ Distribution: This\ is\ a\ statistical\ distribution\ where\ each\ possible\ value\ of\ a\ random\ variable\ is\ equally\ likely. In\ the\ context\ of\ this\ function, it\ means\ that\ each\ possible\ value\ of\ node\ (from\ 1\ to\ the\ length\ of\ the\ list)\ is\ equally\ likely\ to\ be\ requested.$

$Average\ Case: When\ analyzing\ the\ average\ case\ performance\ of\ an\ algorithm, we\ consider\ the\ expected\ performance\ over\ all\ possible\ inputs, assuming\ a\ certain\ distribution\ of\ inputs. For\ the\ getNthNodeFromEnd\ function, if\ we\ assume\ a\ uniform\ distribution\ of\ node, it\ means\ that\ each\ node\ position\ from\ the\ end\ is\ equally\ likely\ to\ be\ requested.$

$Continuing\ with\ the\ calculation:$

$\rightarrow 1st\ we\ have\ to\ calculate\ the\ length\ of\ the\ linked\ list\ i.e.\ `n`, \Theta(n).$

$\rightarrow 2nd\ based\ on\ `node`\ input, where\ `n`\ is\ length\ of\ linked\ list, then:$

$When\ node\ is\ 0, loop\ runs\ `n-1`\ times.$
$When\ node\ is\ 1, loop\ runs\ `n-2`\ times.$
$When\ node\ is\ 2, loop\ runs\ `n-3`\ times.$
.
.
.
$When\ node\ is\ `n-2`, loop\ runs\ 1\ time.$
$When\ node\ is\ `n-1`, loop\ runs\ 0\ time.$

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$$Therefore, when\ node\ entered\ from \left[\sum_{i=0}^{n-1} i\right], loop\ runs\ from \left[\sum_{i=0}^{n-1} i\right] times.$$

$$\sum_{i=0}^{n-1} i = a_0 + \sum_{i=1}^{n-1} i = 0 + \frac{n(n-1)}{2} = \frac{n(n-1)}{2}.$$

$Now, when\ `node = n-1\ or\ len-1`, loop\ runs\ 0\ time\ but\ `current \rightarrow data`\ returns\ i.e.\ the\ first\ node.$

$Hence, we\ have, \frac{n(n-1)}{2} + 1. Here\ 1\ for\ finding\ the\ first\ node.$

$Now, If\ there\ is\ `n`\ no.\ of\ nodes\ there\ we\ find\ data\ at\ `n`\ positions\ equally\ likely\ is:$

$For\ position\ 1, the\ probability\ of\ finding\ data\ is: \frac{1}{n}.$

$For\ position\ 2, the\ probability\ of\ finding\ data\ is: \frac{1}{n}.$

.

.

$For\ position\ n, the\ probability\ of\ finding\ data\ is: \frac{1}{n}.$

$Hence\ probability\ of\ finding\ data\ will\ always\ remain: \frac{1}{n}, equally\ likely.$

$\therefore\ Average\ Time\ Complexity\ is: probability\ of\ finding\ data \times (number\ of\ possible\ iterations + constant\ time\ complexities\ for\ left\ over\ positions).$

$$= \frac{1}{n} \times \left(\frac{n(n-1)}{2} + 1\right)$$

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$$= \frac{n-1}{2} + \frac{1}{n}$$

Appplying Big Theta in the equation:

$$= \Theta\left(\frac{n-1}{2}\right) + \Theta\left(\frac{1}{n}\right)$$

Now, taking out the dominant term of numerator and denominator:

$$= \Theta\left(\frac{n}{2}\right) + \Theta\left(\frac{1}{n}\right)$$

$$= \frac{1}{2} \times \Theta(n) + \Theta\left(\frac{1}{n}\right)$$

Now removing the constant `$\frac{1}{2}$` ***we get***:

$$= \Theta(n) + \Theta\left(\frac{1}{n}\right)$$

When combining two Big Theta Θ **terms , largest growth rate is the result**:

$\rightarrow$ $\Theta(n)$ ***grows linearly with*** `$n$` .

$\rightarrow$ $\Theta\left(\frac{1}{n}\right)$ ***decreases to a constant , i.e. when*** `$n$` ***grows larger ,*** $\frac{1}{n}$ ***approaches to*** 0.

This we can prove by limit and continuity also:

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$$\lim_{n\to\infty}\left(\frac{1}{n}\right) \Rightarrow \frac{1}{\infty} = 0 \left[\frac{c}{\infty} = 0\text{, where } c \text{ is constant}\right].$$

Hence we get `$\Theta(n)$` for all positions.

But as length already calculated earlier i.e. $\Theta(n)$, now calculation for all possibility to get `n` th positions equally likely doesnot actually matters, hence, either from calculation of length of node, we can say directly that Average Case is $\Theta(n)$, or,

Total complexity: $\Theta(n) + \Theta(n) = \Theta(n)$.

Note, by relation: $\Omega \leq \Theta \leq O$, *we get*:

$\Rightarrow c_1 \times g(n) \leq f(n) \leq c_2 \times g(n)$

$\Rightarrow c_1 \times 1 \leq n + \frac{n-1}{2} + \frac{1}{n} \leq c_2 \times n$

Alternatively,

→ *1st we have to calculate the length of the linked list i.e. `n`, $\Theta(n)$.*

For position 1, the probability of finding data is: $\frac{1}{n}$.

For position 2, the probability of finding data is: $\frac{1}{n}$.

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$$For\ position\ N, the\ probability\ of\ finding\ data\ is: \frac{1}{n}.$$

$$Hence\ probability\ of\ finding\ data\ always\ remain\ \frac{1}{n}, for\ position\ 1\ to\ n.$$

$$Average\ case\ complexity:\ Probability\ of\ finding\ data\ \times \begin{pmatrix} Summation \\ of\ number\ of \\ Position. \end{pmatrix}$$

$$= \frac{1}{n} \times \left(\sum_{i=1}^{n} i \right)$$

$$= \frac{1}{n} \times \left(\frac{n(n+1)}{2} \right)$$

$$= \frac{n+1}{2}$$

$$or, \frac{n}{2} + \frac{1}{2}$$

$$Appplying\ Big\ Theta\ in\ the\ equation:$$

$$= \Theta\left(\frac{n}{2}\right) + \Theta\left(\frac{1}{2}\right)$$

$$Now\ removing\ the\ constant\ `\frac{1}{2}`\ we\ get:$$

$$= \Theta(n) + \Theta(1)$$

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$= \Theta(n)$.

$Total\ complexity: \Theta(n) + \Theta(n) = \Theta(n)$.

$Note\ , by\ relation: \Omega \leq \Theta \leq O\ , we\ get:$

$\Longrightarrow c_1 \times g(n) \leq f(n) \leq c_2 \times g(n)$

$\Longrightarrow c_1 \times 1 \leq n + \frac{n+1}{2} \leq c_2 \times n$

Singly Linked List	*Cases*	*Time Complexity*
Get Nth Node From End2	1. *Best Case*	$\Omega(n)$ [*As, length is calculated first hence Worst Case = Best Case , here*]
	2. *Worst Case*	$O(n)$
	3. *Average Case*	$\Theta(n)$

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